Printed Pages: 02 **Sub Code:KCE057**

Paper Id: 231654 **Roll No.**

B.TECH CIVIL ENGINEERING
(SEM V) THEORY EXAMINATION 2022-23
AIR AND NOISE POLLUTION CONTROL

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If you require any missing data, then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2x10 = 20

(a) Define Planetary Boundary layer.
(b) Mention the classification of air pollutants.
(c) What are the features of wind rose.
(d) Define Lapse rate
(e) State Ambient air quality.
(f) Define briefly about emission standards.
(g) List the types of Condensation equipments used in air pollution control.
(h) Define adsorption.
(i) Comment on Noise pollution
(j) State Noise indices.

SECTION B

2. Attempt any *three* of the following: 10x3 = 30

(a) Briefly discuss about the sources and classification of air pollutants.
(b) Explain the types of plume behavior with neat sketch.
(c) Discuss the factors to be considered in selection of equipment for control of particulate contaminants.
(d) Discuss briefly about Euro-I,II,III Specifications.
(e) Explain the control methods and preventive measures undertaken for noise pollution.

SECTION C

3. Attempt any *one* part of the following: 10x1 = 10

(a) Discuss in detail about the effects of air pollution on Human beings, animals and vegetation
(b) Describe in detail about the effects of air pollution on property and aesthetic value and visibility.

4. Attempt any *one* part of the following: 10 x1 = 10

(a) Discuss different types of environmental lapse rate.
(b) Explain the atmospheric factors influencing the dispersion of air pollutants.

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5. Attempt any *one* part of the following: 10x1 = 10

(a) Explain with neat sketches about the principle and working of Cyclone Separator.

(b) Illustrate with neat sketch the working principle of electro static precipitator with its advantages and disadvantages.

6. Attempt any *one* part of the following: 10x1 = 10

(a) Explain various absorption units in air pollution control.

(b) Compare the functions of Central and State Pollution Control Board in the area of air pollution control.

7. Attempt any *one* part of the following: 10x1 = 10

(a) (i) What are the sources of noise? How noise become a pollution problem?

(ii) "Control of noise at source"-Discuss in detail.

(b) Discuss in brief the various sources of noise, and their typical noise levels, in a modern society.

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